

Progress Update Report

Project Title: *Model Distillation for Plant Disease Classification*

Course: Applied Machine Learning (W25)

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1. Progress Made

So far, significant progress has been achieved on the project:

- **Dataset Preparation:** The PlantVillage dataset (Version 3) has been successfully downloaded, preprocessed, and split into training, validation, and test sets.
- **Teacher Model (ResNet152):** The ResNet152 model was successfully finetuned on the dataset using TensorFlow Keras, achieving strong classification performance with over **97% accuracy** on the validation set.
- **Student Model (MobileNetV2):** MobileNetV2 has been initialized and baseline training was conducted without distillation for comparison.
- **Distillation Framework:** A custom knowledge distillation pipeline was implemented, where the MobileNetV2 model learns from the softened outputs (soft targets) of the ResNet152 model using a combined loss function (distillation loss + student loss).

2. Roadblocks and Solutions

- **Issue:** The initial model distillation loss was not decreasing significantly, leading to underwhelming student performance.
Solution: Adjusted the temperature parameter and loss weighting in the distillation process, which helped improve gradient flow and resulted in better learning for MobileNetV2.
- **Issue:** High memory usage during ResNet152 training caused instability on local machines.
Solution: Moved heavy training tasks to cloud-based environments (Free GPU provided by Kaggle) to ensure stable and efficient model training.

3. Next Steps

- **Fine-tune Student Model:** Further optimize MobileNetV2 with hyperparameter tuning and regularization techniques.
- **Model Evaluation:** Compare the performance of the distilled student model with the teacher and baseline models using metrics like accuracy, F1-score, and inference speed.
- **Final Report & Presentation:** Prepare the final documentation, performance analysis, and a short presentation for submission.

This project continues to progress toward delivering a lightweight, accurate, and deployable AI solution for plant disease detection.